

Are Latent Reasoning Tokens Unnecessary for Model Performance?

Anonymous ACL submission

Abstract

Training large language models to perform latent reasoning may improve performance, but the lack of legible reasoning poses challenges for model interpretability and safety assurance. This challenge has motivated interpretability research that aims to characterise the functional role of latent reasoning tokens. Some studies indicate that latent reasoning tokens are effective, whereas other work suggests that these tokens are under-utilised. In this paper, we consider the tension between existing findings by reproducing prior work which shows that some latent reasoning tokens are not necessary. Two interesting observations emerge from our experiments. First, we find that latent reasoning models often generate redundant tokens due to the fixed token budgets. Second, we observe that explicit reasoning models fine-tuned on gold reasoning traces may copy the result of the most recent reasoning step when generating an answer.

1 Introduction

Large language models (LLMs) have become increasingly central to AI research as their capabilities have been improved by scale (Kaplan et al., 2020). These capabilities are especially pronounced in explicit reasoning models (ERMs), such as OpenAI o1 (Jaech et al., 2024) and DeepSeek R1 (Guo et al., 2025), that learn to produce higher quality outputs by starting their responses with a chain-of-thought (CoT): a passage of text in which the model’s given task is broken down into smaller steps that can be tackled one at a time (Wei et al., 2022). Latent reasoning models (LRMs), such as COCONUT (Hao et al., 2024) and CODI (Shen et al., 2025), replace these human-readable steps with an illegible array of numbers, as shown in Figure 1. This modification may improve reasoning performance, but it also raises new challenges for our understanding of these systems.

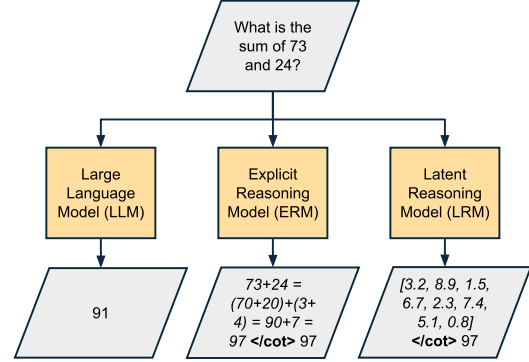


Figure 1: A comparison of model outputs.

Interpreting LLMs is difficult due to complications like polysemanticity and superposition (Olah et al., 2020; Elhage et al., 2022). However, explicit CoTs have proved advantageous, as they can provide observers with information about model reasoning. This information can be leveraged to improve model safeguards, such as monitoring systems, as shown by Baker et al. (2025). While latent reasoning models may improve performance, they also raise new challenges for model interpretability and safety assurance (Korbak et al., 2025).

Work on interpreting latent reasoning models has been motivated by a need to address these risks. Some research has attempted to decode latent reasoning tokens (Shen et al., 2025), or replace them with a legible representations (Wang et al., 2026). Hao et al. (2024) show that COCONUT uses its latent CoT to attempt multiple solutions in parallel, while Goyal et al. (2025) show that CODI’s confidence in the correct answer gradually increases as additional latent reasoning tokens are generated. In contrast, Dilgren and Wiegrefe (2026) show that some LRMs may determine their final answer before generating part or all of their CoT. In this paper, we aim to address the tension between these findings and characterise the functional role of latent reasoning tokens.

The rest of the paper is structured as follows, in Section 3 we replicate the experiment performed by Dilgren and Wiegrefe (2026). We proceed by performing ablations to understand the factors contributing to the discrepancy between the results reported for COCONUT, CODI, and the ERM. We identify two factors that contribute to this discrepancy: the first is the fixed token budgets used by the LRMs (Section 4), which force them to generate redundant latent reasoning tokens; the second is the tendency of the ERM to copy the result of the most recent explicit reasoning step when generating an answer (Section 5), which inflates perceived reasoning step necessity since truncating the CoT often causes the ERM to change its answer.

2 Background

Latent Reasoning Models: Latent reasoning models work by applying recurrent computations to the final hidden state. Two distinct categories of latent reasoning models can be identified: vertical recurrence models, such as Huginn-0125 (Geiping et al., 2025) and Ouro (Zhu et al., 2025), that repeatedly apply intermediate layers in a deep neural network to the final hidden state; and horizontal recurrence models, such as Switch (Yang et al., 2026) and SIM-CoT (Wei et al., 2025), that generate latent reasoning through an autoregressive process where the final hidden state is combined with the prompt and used as input to another forward pass. In this paper, we focus on two horizontal-recurrence-based latent reasoning models: COCONUT (Hao et al., 2024) and CODI (Shen et al., 2025).

Latent Reasoning Interpretability: Existing work has commented on whether or not latent reasoning tokens are necessary for model performance, but conclusions appear to be mixed. Wang et al. (2026) show that features in sparse latent reasoning tokens are causally effective operators. Li et al. (2026) find that some latent reasoning tokens have an outsized causal influence on the model’s final answer, while some aren’t as effective, and that a correct answer can often be decoded early. Similarly, Aswal et al. (2026) also show that the causal effect of latent reasoning tokens on the model’s final answer is varied, and Chang et al. (2026) find that early latent reasoning tokens act as critical causal hubs. Goyal et al. (2025) investigate a mechanism that may explain the heterogeneity in CODI.

It is shown that interventions on even latent reasoning tokens, which appear to store information, are more impactful than interventions on odd latent reasoning tokens, which appear to be transitional states produced during computation over that stored information. Cui et al. (2026) report pervasive "shortcut" behaviour, where LRMs perform well without utilising their latent reasoning tokens. Viveiros et al. (2026) examines visual LRMs and produce similar results, finding that performance wasn’t significantly different when reasoning tokens were swapped out for dummy tokens. Jin et al. (2026) show that the necessity of latent reasoning tokens can change during the course of training. Our work is particularly focused on Dilgren and Wiegrefe (2026), who find that truncating the latent CoT often doesn’t change the models answer, and conclude that latent reasoning tokens may not be necessary for model performance.

3 Reproduction

We reproduce the experimental setup described by Dilgren and Wiegrefe (2026), which aims to determine the extent to which reasoning tokens are necessary for model performance. Problems are sampled from a dataset and given to a reasoning model, which generates a response to each problem. A model’s response is comprised of a CoT, which consists of reasoning tokens, and the model’s final answer. Several truncated versions of the CoT are created by removing all reasoning tokens after a given point. The model is prefilled with the problem statement and each truncated CoT, and then forced to generate an answer as we terminate the CoT by inserting an <eot> token. The resulting answers are compared to the model’s original answer to determine the impact of the removed reasoning. In the rest of this section we provide additional details on the experimental setup.

Datasets: We use the same 3 datasets as Dilgren and Wiegrefe (2026): GSM8k (Cobbe et al., 2021), ProsQA (Hao et al., 2024), and PrOntoQA (Saparov and He, 2022). GSM8k consists of numerical math problems solvable by "a bright middle school student", whereas both ProsQA and PrOntoQA consist of logic problems. Each GSM8k sample consists of a problem statement, and a gold reasoning trace. This gold reasoning trace consists of a natural language worked solution to the associated problem. Where numerical operations are described in natural language, they are accom-

Model	GSM8k		ProsQA		PrOntoQA	
	Ours	D&W	Ours	D&W	Ours	D&W
ERM	41.5%	41.6%	74.2%	74.2%	99.2%	99.3%
COCONUT	30.5%	33.1%	93.8%	98.0%	98.9%	99.0%
CODI	41.6%	42.2%	81.4%	81.6%	94.9%	95.1%

Table 1: **Testing accuracy of explicit reasoning model and two latent reasoning models (COCONUT, CODI) on three datasets.** D & W defines the corresponding score reported by [Dilgren and Wiegrefe \(2026\)](#).

panied by a representation of those operations in calculator notation, and the worked solution ends with a correct answer. We split GSM8k into a training split comprised of 7473 samples, and a test split comprised of 1319 samples.

Models: We evaluate the same three models used by [Dilgren and Wiegrefe \(2026\)](#).¹ Each of the three models is created by applying a different training procedure to GPT-2 small ([Radford et al., 2019](#)). The first model has explicit reasoning capabilities, and it is created by performing supervised fine-tuning on gold reasoning traces. These reasoning traces are taken from GSM8k-Aug ([Deng et al., 2023](#)), which integrates additional samples into the GSM8k training split, and is preprocessed by removing all natural language prose. The remaining models have latent reasoning capabilities. Specifically, the second model is created by applying the COCONUT training procedure described by [Hao et al. \(2024\)](#), and the third model is created by applying the CODI training procedure outlined by ([Shen et al., 2025](#)). Our models have similar performance to those in [Dilgren and Wiegrefe \(2026\)](#), as shown in Table 1.

Metrics: We compute the two metrics described by [Dilgren and Wiegrefe \(2026\)](#). The first is *first match*, which is equal to the minimum number of reasoning tokens that must be generated before a model first produces the same answer as it would have if it had finished generating its CoT. The second metric is *stable match*, which is equal to the minimum number of reasoning tokens after which the model does not change its answer regardless of the number of additional tokens remaining. In line with [Dilgren and Wiegrefe \(2026\)](#), we divide first match and stable match by the number of latent reasoning tokens or explicit reasoning steps in the corresponding CoT and report the fraction as the

result. In Figure 2 we show that our reproduction achieves similar matching metrics as [Dilgren and Wiegrefe \(2026\)](#).

4 Fixed Token Budgets Produce Redundant Reasoning

In this section we investigate whether the observed gap in matching metrics between the LRMs and ERMs is caused by the LRMs’ fixed latent reasoning token budgets and the ERMs’ variable CoT length. This could be the case if the ERM is able to terminate its CoT early on simple problems, where the LRMs would be forced to generate redundant latent reasoning tokens in accordance with their fixed budget.

COCONUT and CODI do indeed generate more redundant tokens on simpler problems, while the ERM’s matching metrics are unaffected by problem complexity. We bucket GSM8k samples by the number of computational steps represented with calculator notation in the gold reasoning trace, and use this number of steps as a proxy for problem complexity. Figure 3 shows that as the number of steps in a sample’s gold reasoning trace increases, the LRMs’ stable match fraction generally also increases. On the other hand, ?? shows that the ERM’s stable match remains close to saturated, regardless of the number of steps in a sample’s gold reasoning trace.

We also find that COCONUT, CODI, and the ERM all generate a similar number of effective reasoning steps, as shown in Figure 4. Here, we define effective reasoning steps to be the number of latent reasoning tokens or explicit reasoning steps that are generated before the model converges on its final answer. Specifically, here the effective reasoning step is defined as the stable match fraction (Section 3) multiplied by the total number of reasoning steps the model generates per question – a fixed 6 latent thoughts for COCONUT/CODI, or the model’s own variable number of explicit reasoning steps for the ERM – recovering the number of

¹We use the same checkpoints as [Dilgren and Wiegrefe \(2026\)](#): <https://huggingface.co/collections/connordilgren/are-latent-reasoning-models-easily-interpretable> and <https://huggingface.co/zen-E/CODI-gpt2>

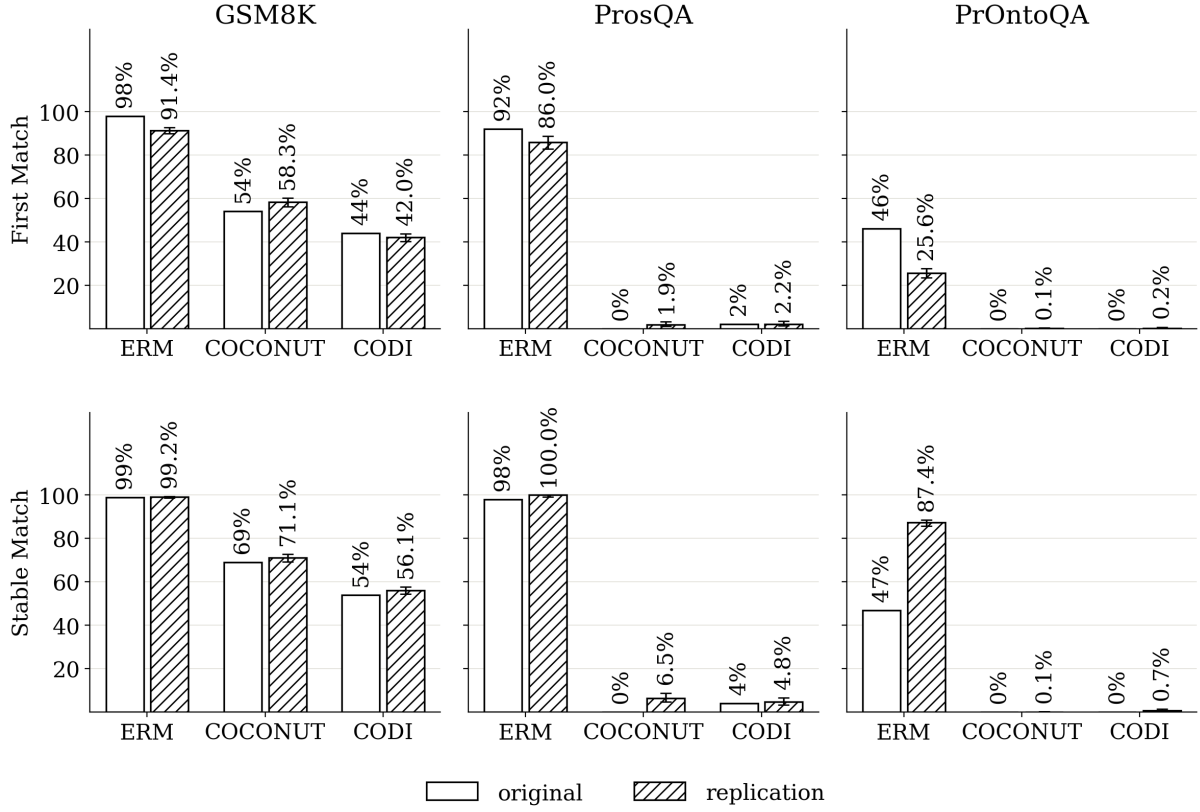


Figure 2: **Our reproduction produces similar *first match* and *stable match* scores as Dilgren and Wiegrefe (2026).** Error bars on the reproduction bars are a 95% bias-corrected and accelerated (BCa) bootstrap confidence interval (10,000 resamples) over the per-question first/stable match fractions; no comparable interval is available for the originally reported numbers, since only a single summary figure was published for each.

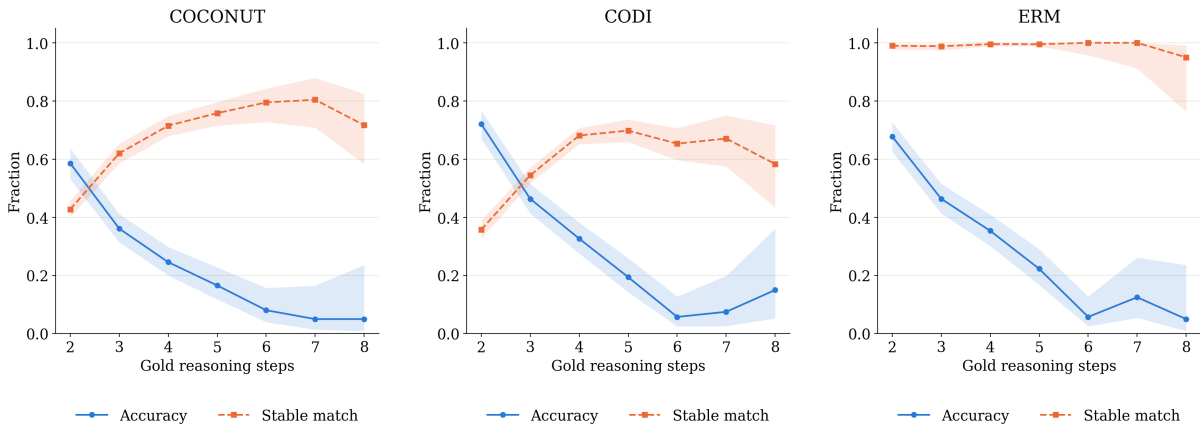


Figure 3: **Testing accuracy and stable match score with increasing gold reasoning steps (GSM8k).** Shaded region is 95% confidence interval: a Wilson score interval for accuracy, since it is a binomial correct/incorrect proportion; and a BCa bootstrap confidence interval (10,000 resamples) over per-question stable match fractions for stable match. As number of steps increases task difficulty is increased, as shown by reduced performance, and, in COCONUT and CODI, stable match increases.

steps actually needed before the answer stabilizes, on a scale comparable across models with different step budgets. This similarity in effective reasoning steps suggests that the difference in matching metrics is driven by redundant latent reasoning tokens that COCONUT and CODI are forced to generate in order to fill their fixed budget.

Dilgren and Wiegrefe (2026) propose two hypotheses that may explain why COCONUT and CODI produce lower matching metrics than the ERM: the first is that low task difficulty may cause matching metrics to be low in COCONUT and CODI, but not the ERM; the second is that COCONUT and CODI under-utilise their reasoning tokens relative to the ERM. Our results in this section support the first hypothesis, since we find that the LRM’s matching metrics are affected significantly by task difficulty, *i.e.*, they are lower on less difficult tasks, while the ERM’s matching metrics are not, as shown in Figure 3 and ??.

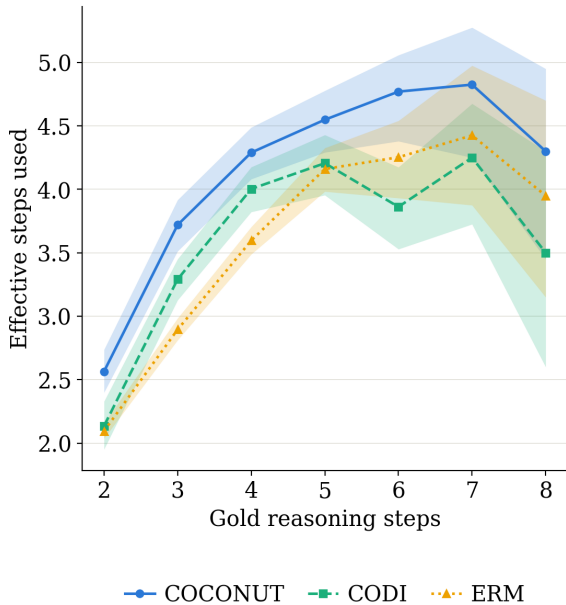


Figure 4: **Effective reasoning steps of three models with increasing gold reasoning steps (GSM8k).** All three models emit a similar number of effective reasoning steps on GSM8k questions with a given number of steps in the gold reasoning trace. Shaded regions are a 95% BCa bootstrap confidence interval (10,000 resamples) over per-question effective-step values.

5 ERM’s Answers May Be Copied From the Last Reasoning Step

In this section we investigate whether the ERM has a tendency to copy the result of the last reason-

ing step when generating answers. This behaviour could account for some of the discrepancy in matching metrics between the LRMs and the ERM, since it could cause the ERM to change its answer each time the CoT is truncated.

We find that 64% of the answers generated by the ERM while running the early stopping experiment on GSM8k were the same as the result of the preceding explicit reasoning step. We note that this includes answers that were generated while the CoT was truncated. We also run a variation of the early stopping experiment: before the model generates an answer we swap the result of the last explicit reasoning step with a random number within the same order of magnitude. When running this variation, the ERM’s final answer matches this injected random number in 45.9% of cases. It should be noted that in most cases, the random number has no plausible connection to the question. These results suggest that the ERM may have a tendency to copy the result of the last explicit reasoning reasoning step when generating an answer. However, this bias would only increase ERM’s matching metrics relative to those of the LRMs if the LRMs did not have similar biases.

To determine whether or not the LRMs have similar biases, we conduct the following experiment: for each GSM8k sample, we create an augmented version by replacing the first number in the problem statement with a random number within the same order of magnitude. We run each model on each augmented GSM8k sample, and record the resulting explicit reasoning steps and latent reasoning tokens. Then, we run the early stopping experiment on the original GSM8k samples. However, before each model generates an answer, we swap the last latent reasoning token or explicit reasoning step with the corresponding token or step produced by that model on the corresponding augmented question. We exclude samples where: 1) a different number of explicit reasoning steps were produced for the original question and the augmented question, 2) the model produced the same answer for both questions, and 3) the question did not contain an number that could be perturbed. Answers are categorised as *augmented*, if the swap causes the model’s answer to match the answer given to the augmented question, *original* if the swap resulted in no change, or *other* if the resulting answer no longer matched either the original answer or the augmented one. In Figure 5 we see that the ERM changes its answer significantly more often than

either LRM, suggesting that the LRMs do not have an analogous copying bias to the ERM.

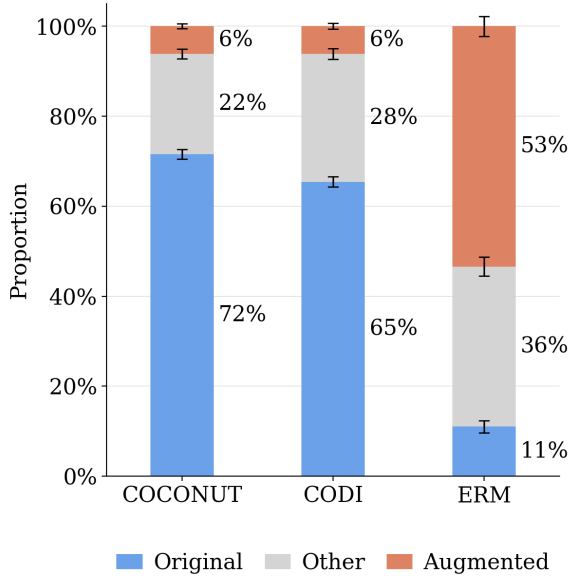


Figure 5: Answer status after splicing one augmented-question reasoning step into an otherwise-original trace (GSM8k). The resulting answer either still matches the original question (*original*), switches to match the augmented question (*augmented*), matches neither (*other*). The ERM switches to the augmented answer far more often than COCONUT or CODI, consistent with a copying bias. Error bars are a 95% cluster bootstrap confidence interval (10,000 resamples): whole questions, rather than individual steps, are resampled with replacement, since steps from the same question are correlated rather than independent trials.

6 Conclusion

In this paper, we have successfully reproduce the results of Dilgren and Wiegrefe (2026), and designed targeted experiments to identify two factors that may explain the gap in matching metrics between the ERM, and the LRMs (COCONUT and CODI). First, fixed token budgets force latent reasoning models to generate redundant tokens, thereby reducing their matching metrics relative to the ERM. Second, the ERM has a tendency to copy the result of the most recent explicit reasoning step when generating its final answer, thereby increasing its matching metrics relative to the LRMs. Our results characterise the functional role of latent reasoning tokens in LRMs and contribute to a better understanding of the interpretability of latent reasoning models.

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